

KINETIC COMPARISON OF THE POWER DEVELOPMENT BETWEEN POWER CLEAN VARIATIONS

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ABSTRACT

Suchomel, TJ, Wright, GA, Kernozek, TW, and Kline, DE. Kinetic comparison of the power development between power clean variations. *J Strength Cond Res* 28(2): 350–360, 2014—The purpose of this study was to compare the power production of the hang clean (HC), jump shrug (JS), and high pull (HP) when performed at different relative loads. Seventeen men with previous HC training experience, performed 3 repetitions each of the HC, JS, and HP at relative loads of 30, 45, 65, and 80% of their 1 repetition maximum (1RM) HC on a force platform over 3 different testing sessions. Peak power output (PPO), peak force (PF), and peak velocity (PV) of the lifter plus bar system during each repetition were compared. The JS produced a greater PPO, PF, and PV than both the HC ($p < 0.001$) and HP ($p < 0.001$). The HP also produced a greater PPO ($p < 0.01$) and PV ($p < 0.001$) than the HC. Peak power output, PF, and PV occurred at 45, 65, and 30% 1RM, respectively. Peak power output at 45% 1RM was greater than PPO at 65% ($p = 0.043$) and 80% 1RM ($p = 0.004$). Peak force at 30% was less than PF at 45% ($p = 0.006$), 65% ($p < 0.001$), and 80% 1RM ($p = 0.003$). Peak velocity at 30 and 45% was greater than PV at 65% ($p < 0.001$) and 80% 1RM ($p < 0.001$). Peak velocity at 65% 1RM was also greater than PV at 80% 1RM ($p < 0.001$). When designing resistance training programs, practitioners should consider implementing the JS and HP. To optimize PPO, loads of approximately 30 and 45% 1RM HC are recommended for the JS and HP, respectively.

KEY WORDS hang clean, jump shrug, high pull, resistance training

INTRODUCTION

It has been well documented that a strong relationship exists between the ability of an athlete to develop high levels of muscular power and their success in sports (1,3,4,6,7,10,11,13–15,18,20,21,23–26,28–31). Common movements in sports, such as sprinting and jumping, require an athlete to produce high amounts of power. Furthermore, many coaches and researchers believe that the ideal stimulus for improving muscular power is by utilizing methods that produce maximal power in sports-specific movements (24). Thus, the development of lower body muscular power is a primary focus of many strength and conditioning practitioners in their training programs for the purpose of improving the overall performance of athletes in sports.

The power clean and its variations are commonly used to train lower body muscular power (2,3,9,14–16,18–21, 23–25,28,30,31). By implementing the power clean and its variations into training programs, strength and conditioning practitioners train lower body power, highlighted by the explosive extension of the hip, knee, and ankle joints (23). This is commonly referred to as triple extension. The second pull phase, which produces the greatest ground reaction force and power output during Olympic style lifts (11–13,20), is similar to many sport movements and therefore, is the phase that classifies the power clean and its variations as sports specific (20). To emphasize power development during the second pull phase, practitioners often prescribe power clean variations that are performed from the hang position. Of these hang variations, the hang clean (HC) may be the most commonly used. Despite being a highly beneficial exercise, the HC may be more time consuming to teach an athlete as compared with lead-up exercises used to teach the HC (16). It has been suggested that it may be essential to substitute less technical exercises to train lower body muscular power (1,20). This raises the question: are power clean variations that are used to teach the HC as effective at producing lower body muscular power as the HC itself?

Two variations used to teach the power clean are the jump shrug (JS) and high pull (HP) (16,21). Similar to the HC, the JS and HP can both be performed from the hang position and are used to train lower body power. However, only 1 study has examined the HP (32), whereas no previous literature has

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28(2)/350–360

Journal of Strength and Conditioning Research
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TABLE 1. Activities performed during each session during the study.*

Familiarization session	Testing sessions (2–7 days after previous trial)
Subjects read and signed informed consent	Performed dynamic warm-up
Performed dynamic warm-up	Performed 3 maximal effort repetitions of one of the 3 exercises (HC, JS, HP) performed at each of the loads (30, 45, 65, and 80% 1 RM HC)
1RM HC testing	Exercises performed in randomized order
Familiarization of JS and HP exercises	

*HC = hang clean; JS = jump shrug; HP = high pull; 1RM = 1 repetition maximum.

examined the extent to which the JS exercise can train lower body muscular power. Thomas et al. (32) compared a free-form HP to a fixed-form HP across a loading spectrum ranging from 30 to 70% 1 repetition maximum (1RM). The authors displayed no significant interaction difference regarding the form of the HP and that maximal power output occurred between 30 and 60% 1RM. Although the above results contribute to the literature regarding power clean variations, it is still unknown how the JS and HP exercises compare to the power produced by the HC. If strength and conditioning practitioners are looking for an alternative lower body exercise that trains lower body power, there is a need to investigate the potential lower body power development of the JS and HP. Furthermore, it is important that the differences in power development between the HC, JS, and HP are presented to strength and conditioning practitioners so that they can choose an exercise that will allow their athletes to train lower body power effectively.

It has been suggested that strength and conditioning practitioners should select exercises that maximize power output during the movement that is being trained (28). Furthermore, by identifying and training with the ideal load for a specific exercise, athletes will be able to optimally improve their muscular power and as a result, their overall athletic performance (22,32). Therefore, the purpose of this study is to compare the power production of the HC, JS, and HP when performed at different loads relative to the 1RM HC of each subject. Based on previous pilot testing and the ballistic nature of the JS, it was hypothesized that the JS would produce the greatest power output as compared with the HC and HP exercises. Previous research has indicated that the optimal load for the HC and power clean exercises occurred at either 70% (4,23) or 80% 1RM (6,9,10,25). However, it should be noted that several studies observed that there was no significant difference between the optimal load and 60–80% 1RM (4) or 50–90% 1RM (9,23,25). Based on previous research and pilot testing, it was hypothesized that the greatest power output for the HC would occur at 65% 1RM HC.

METHODS

Experimental Approach to the Problem

To test our hypotheses, a repeated measures design was used to investigate the relationships between power clean variations (HC, JS, and HP) performed at different relative loads (30, 45, 65, and 80% 1RM HC) and the peak power output (PPO) produced during the repetitions of each exercise and load. The exercises (HC, JS, and HP) and loads (30, 45, 65, and 80% 1RM HC) were chosen as independent variables to compare the kinetics produced during each repetition to determine if any differences existed between the respective exercises and loads. The specific relative loads were chosen to cover a wide range of light, moderate, and heavy training loads. The PPO, peak force (PF), and peak velocity (PV) of each repetition were chosen as dependent variables because they are frequently compared when investigating the power clean and its variations (2–6,9,11,15,18,19,23,25,33). Because power is a product of force and velocity, it was deemed necessary to examine the factors contributing to power production. Subjects completed a single familiarization session and 3 different testing sessions. Testing sessions for each subject were completed at the same time of day and were separated by minimum of 2 days and maximum of

TABLE 2. Subject demographics and performance characteristics: $n = 17$.

Variable	Mean $\pm$ SD
Age (y)	21.59 $\pm$ 1.28
Height (cm)	180.87 $\pm$ 6.27
Body mass (kg)	87.13 $\pm$ 15.60
1RM hang clean (kg)	111.12 $\pm$ 20.40
Hang clean training experience (y)	3.91 $\pm$ 1.23



Figure 1. Barbell lowered to the countermovement hang position from the starting position.



Figure 2. Finishing position of the hang clean.

7 days between sessions. The familiarization session was used to obtain the subject's 1RM HC and to familiarize the subjects with the JS and HP exercises. During each testing session, subjects completed randomized sets of one of the exercises (HC, JS, or HP) on a force platform although the vertical ground reaction forces at different relative loads were collected. The activities performed during each session are displayed in Table 1.

Subjects

Seventeen athletic males with a minimum of 2 years of previous training experience with the HC exercise, but no previous competitive weightlifting experience, agreed to participate in the present study. Each subject competed in Division III collegiate track and field (short sprints, jumps, or throws) ($n = 8$) or collegiate club/intramural sports ($n = 9$). All subjects were tested during the same time of year, which coincided with the offseason portion of the track and field training program. The age, height, body mass, 1RM HC, and HC training experience of the subjects are listed below in



Figure 3. Finishing position of the jump shrug.



Figure 4. Finishing position of the high pull.

Table 2. Subjects were asked to refrain from physical activity that may affect testing performance, the consumption of alcohol, caffeine, and other ergogenic aids at least 24 hours before each testing session. If subjects did not meet these

standards upon their arrival, they were asked to reschedule their testing session within the 2–7 days window previously described. This study was approved by the University of Wisconsin-La Crosse Institutional Review Board. All subjects were informed of the possible risks of involvement in the study and provided written informed consent.

Procedures

One Repetition Maximum Hang Clean Testing. Each subject's 1RM HC was determined by using the protocol previously established by Winchester et al. (33) using the HC technique previously described by Kawamori et al. (23). Before performing any maximal HC attempts, each subject completed a standardized dynamic warm-up (e.g., stationary cycling, lunges, countermovement jumps, etc.) lasting approximately 8 minutes. After the dynamic warm-up, subjects performed several submaximal sets of the HC exercise (e.g., 30, 50, 70, 90% of estimated 1RM HC) as part of the 1RM HC warm-up. Briefly, the HC exercise started from a standing position with the subject holding the bar using an overhand grip. Subjects then lowered the bar down their thighs to just above knee level (Figure 1), lifted the bar explosively upward, and caught the bar across their shoulders in a semi-squat position (Figure 2). The HC repetition was termed unsuccessful if the researcher observed that the subject's upper thigh fell below parallel to the floor during the catch phase (33). After the subject's 1RM HC was established, subjects were familiarized with the technique of the JS and HP exercises. The JS and HP required the subject to start in a standing position and lower the bar down their thighs until the bar was just above their knees, identical to the beginning of the HC. The JS required the subject to maximally jump with the barbell although violently shrugging their shoulders

(12,16,21). A successful repetition of the JS required the subject to leave the surface of the force platform (Figure 3). After the bar was lowered to a position just above their knees, as described above, the HP required the subject to explosively extend their hips, knees, and ankles; shrug their shoulders; drive their elbows upward; and elevate the barbell to chest height (12,21) (Figure 4). A successful HP repetition was determined if the subject lifted the bar explosively upward and elevated the barbell to chest height.

TABLE 3. ICC, CV percent, and 90% confidence interval ranges of the measurements: $n = 17$.*

Exercise and variable	ICC range	CV % range	90% Confidence interval	
			Lower bound	Upper bound
Hang clean				
Power	0.88–0.96	8.3–16.9	6.7–13.7	11.2–23.1
Force	0.95–0.99	3.5–7.1	2.8–5.8	4.7–9.5
Velocity	0.87–0.94	6.6–10.1	5.4–8.3	9.0–13.7
Jump shrug				
Power	0.93–0.98	5.1–8.2	4.0–6.7	8.4–11.0
Force	0.99–0.99	2.2–2.9	1.8–2.3	2.9–3.8
Velocity	0.76–0.84	4.2–7.2	3.3–5.9	5.9–9.7
High pull				
Power	0.96–0.98	6.7–9.5	5.4–7.8	9.2–13.0
Force	0.98–0.99	3.7–4.0	3.0–3.2	5.0–5.5
Velocity	0.91–0.95	4.8–7.5	3.9–6.1	6.5–10.4

*ICC = intraclass correlation coefficient; CV = coefficient of variation.

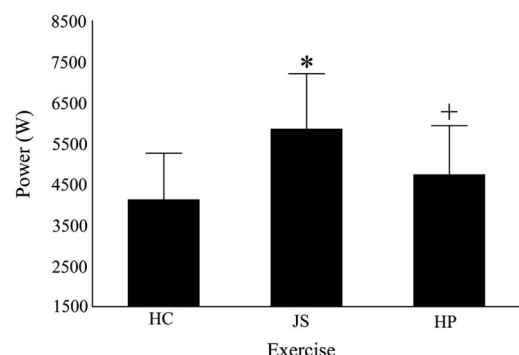


Figure 5. Exercise main effects for peak power output. *Significantly greater than HC and HP ($p < 0.001$); +significantly greater than HC ($p = 0.001$). HC = hang clean; JS = jump shrug; HP = high pull.

Power Testing. The dynamic warm-up and light exercise sets (e.g., 30, 50% 1RM HC) of the exercise that was to be performed that day (HC, JS, or HP) were performed before testing. For example, if the subjects were performing the JS during a testing session, they would perform sets of the JS at 30 and 50% 1RM HC as a part of their warm-up before performing testing repetitions. Subjects completed 3 maximal effort repetitions at each of their relative loads (30, 45, 65, and 80% of their 1RM HC) in a randomized order using the exercise of the day. Therefore, each testing session required the subject to perform 12 total repetitions. The same randomized order of relative loads was used during each testing session with the different exercises. In addition, the relative loads remained constant between each exercise. Due to the large number of repetitions, only 1 exercise (HC, JS, or HP) was tested per visit in a randomized order to

prevent fatigue. Sixty seconds of rest was provided between each repetition (15), whereas 2 minutes were provided between each load. The bar was placed on the safety bars of a squat rack in between all repetitions to prevent fatigue. All repetitions of each exercise were performed on a portable Kistler Quattro Jump force platform (Type 9290AD; Kistler, Winterthur, Switzerland) interfaced with a laptop computer and were sampled at 500 Hz. The methodology of using a force platform only apparatus during weightlifting exercises is supported by Hori et al. (18,19). Finally, subjects were encouraged to complete each repetition with maximal effort.

Data Analysis

Peak power output, PF, and PV of the center of mass of the lifter plus bar system were calculated from the vertical ground reaction forces of the HC, JS, and HP using a template created in Microsoft Excel (Microsoft Corporation, Redmond, VA, USA). The greatest PPO, PF, and PV values produced by each subject during the HC, JS, and HP at each load were used for comparison. Vertical ground reaction forces of the lifter plus bar system were measured directly with the force platform. The velocity and power output of the center of mass of the lifter plus bar system were calculated using a forward dynamics approach previously established (23).

Statistical Analyses

All data are reported as the mean $\pm$ SD. A series of 3 (exercise) $\times$ 4 (load) repeated measures analysis of variance were used to compare the main effect differences of the PPO, PF, and PV produced between the HC, JS, and HP exercises and the various loads (30, 45, 65, 80% 1RM HC). When necessary, post hoc analyses were performed using the Bonferroni technique. All statistical analysis was performed using SPSS 20.1 (IBM, New York, NY, USA). For all statistical tests, the alpha value was set at 0.05. Statistical power was calculated between 0.87 and 1.00 for all measures. Effect sizes were

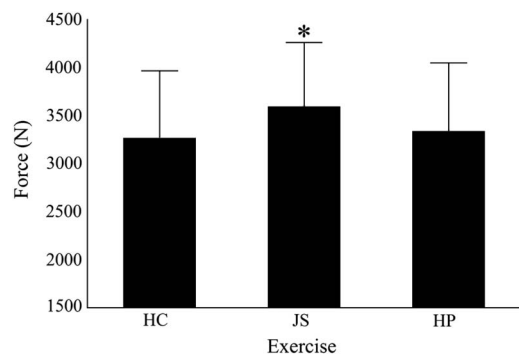


Figure 6. Exercise main effects for peak force. *Significantly greater than HC and HP ($p < 0.001$); HC = hang clean; JS = jump shrug; HP = high pull.

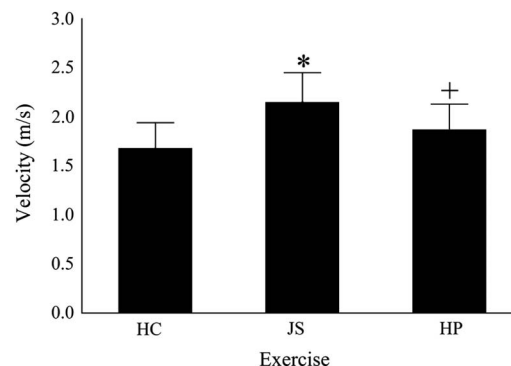
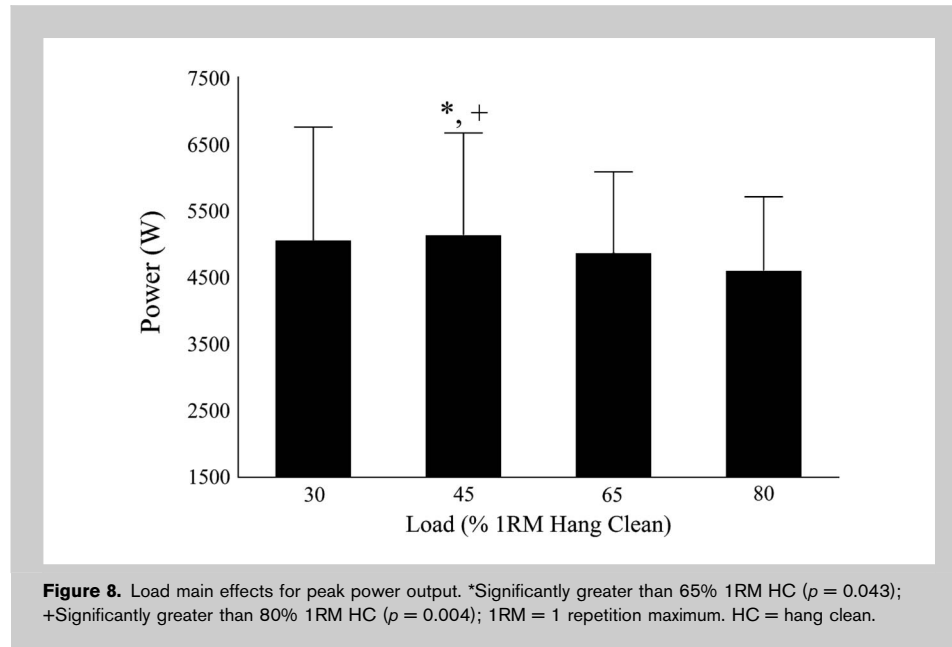
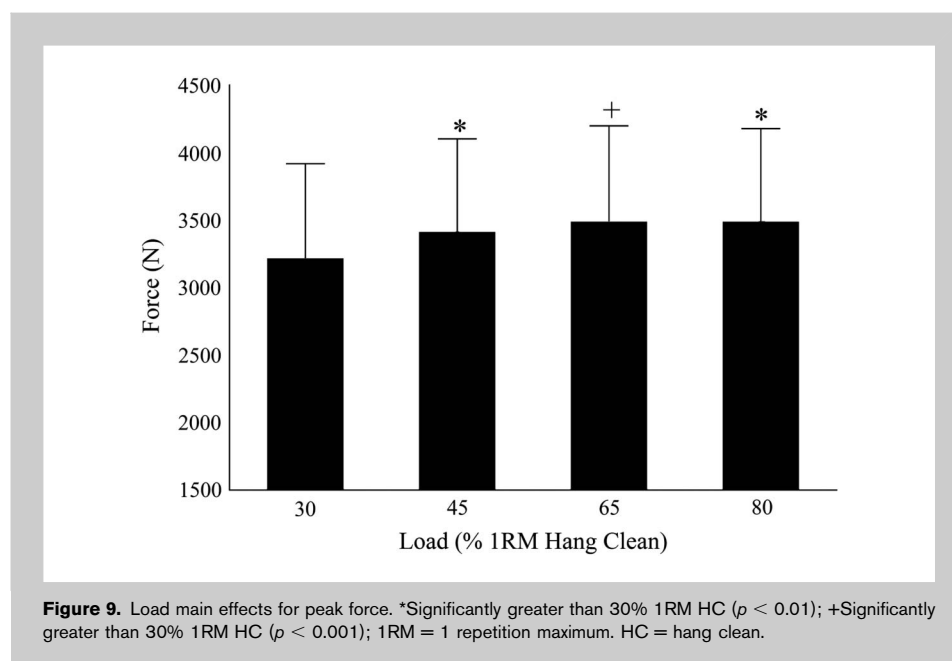


Figure 7. Exercise main effects for peak velocity. *Significantly greater than HC and HP ($p < 0.001$); +Significantly greater than HC ($p < 0.001$); HC = hang clean; JS = jump shrug; HP = high pull.



calculated using Cohen's d and were interpreted using the scale developed by Hopkins (17), where effect sizes were considered trivial, small, moderate, large, very large, and nearly perfect when Cohen's d was 0.0, 0.2, 0.6, 1.2, 2.0, and 4.0, respectively. Finally, intraclass correlation coefficients, coefficients of variation, and 90% confidence interval ranges were calculated from the 3 separate repetitions of each exercise at each load and are displayed in Table 3. Ranges displayed indicate the values found at each load for each variable.



RESULTS

Exercise

Exercise PPO main effect results are displayed in Figure 5. Significant differences in PPO occurred between the HC, JS, and HP exercises ($p < 0.001$). Post hoc analysis revealed a significantly greater PPO during the JS (5851.38 ± 1354.94 W) compared with both the HC (4123.61 ± 1135.32 W) ($p < 0.001$, $d = 1.38$) and HP (4737.08 ± 1196.36 W) ($p < 0.001$, $d = 0.87$). In addition, the PPO of the HP was significantly greater than the HC variation ($p = 0.001$, $d = 0.53$).

Exercise PF main effect results are displayed in Figure 6. Significant differences in PF were identified between the

HC, JS, and HP exercises ($p < 0.001$). Post hoc analysis revealed a significantly greater PF during the JS (3593.99 ± 666.20 N) compared with both the HC (3267.19 ± 698.16 N) ($p < 0.001$, $d = 0.48$) and the HP (3337.02 ± 710.46 N) ($p < 0.001$, $d = 0.37$). However, no significant difference in PF existed between the HC and HP variations of the power clean ($p = 0.309$, $d = 0.10$).

Exercise PV main effect results are displayed in Figure 7. Significant differences in PV occurred between the HC, JS, and HP exercises ($p < 0.001$). Post hoc analysis revealed

a significantly greater PV during the JS (2.15 ± 0.30 m $\cdot$ s $^{-1}$) compared with both the HC (1.68 ± 0.26 m $\cdot$ s $^{-1}$) ($p < 0.001$, $d = 1.67$) and HP (1.87 ± 0.26 m $\cdot$ s $^{-1}$) ($p < 0.001$, $d = 1.00$). In addition, the PV of the HP was significantly greater than the HC variation ($p < 0.001$, $d = 0.73$).

Load

Load PPO main effects are displayed in Figure 8. Significant main effects in PPO occurred between different loads during the HC, JS, and HP exercises ($p < 0.001$). The greatest PPO occurred at 45% 1RM HC (5124.82 ± 1538.26 W). This was followed in order by 30% (5045.81 ± 1705.51

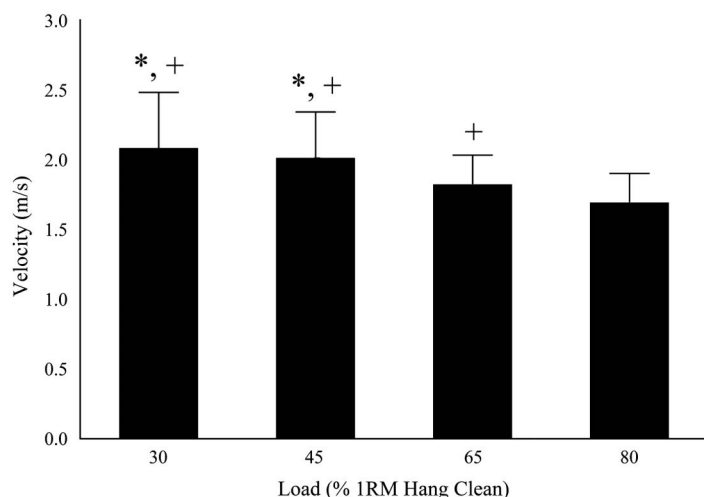


Figure 10. Load main effects for peak velocity. *Significantly greater than 65% 1RM HC ($p < 0.001$); +significantly greater than 80% 1RM HC ($p < 0.001$); 1RM = 1 repetition maximum. HC = hang clean.

W), 65% (4854.31 ± 1224.85 W), and 80% 1RM HC (4591.15 ± 1115.38 W). Post hoc analysis revealed that the PPO at 45% 1RM HC was significantly greater than PPO that occurred at 65% ($p = 0.043$, $d = 0.19$) and 80% 1RM HC ($p = 0.004$, $d = 0.40$). However, PPO at 45% 1RM HC was not significantly different from PPO at 30% 1RM HC ($p = 1.000$, $d = 0.05$). No other significant differences in PPO occurred between 30 and 65% 1RM HC ($p = 0.663$, $d = 0.13$), 30 and 80% 1RM HC ($p = 0.105$, $d = 0.32$), or 65 and 80% 1RM HC ($p = 0.134$, $d = 0.22$).

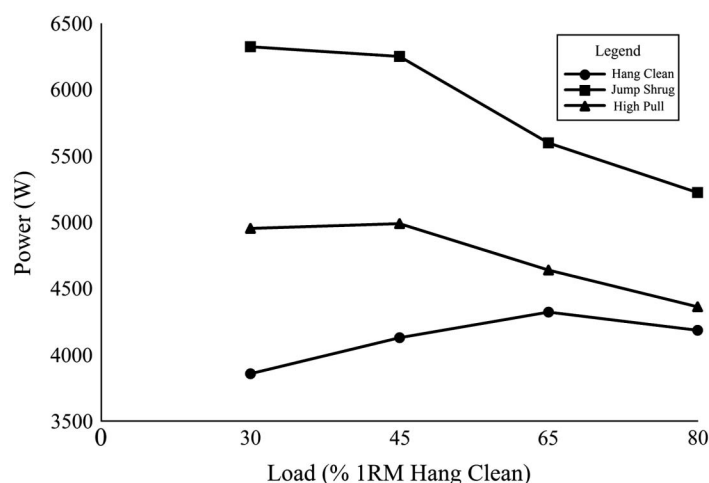


Figure 11. Exercise and load interaction for peak power output ($p < 0.001$). 1RM = 1 repetition maximum.

Load PF main effects are displayed in Figure 9. Significant differences in PF were observed between the different loads during the HC, JS, and HP exercises ($p < 0.001$). The load of 65% 1RM HC displayed the highest PF (3487.15 ± 710.75 N). This was followed in order by 80% (3485.96 ± 690.90 N), 45% (3409.71 ± 692.19 N), and finally 30% 1RM HC (3214.78 ± 701.99 N). Post hoc analysis revealed that the exercise load of 30% 1RM HC yielded significantly lower PF than 45% ($p = 0.006$, $d = 0.28$), 65% ($p < 0.001$, $d = 0.39$), and 80% 1RM HC ($p = 0.003$, $d = 0.39$). However, no significant differences in PF existed between 45 and 65% 1RM

HC ($p = 0.319$, $d = 0.11$), 45 and 80% 1RM HC ($p = 0.427$, $d = 0.11$), or 65 and 80% 1RM HC ($p = 1.000$, $d = 0.00$).

Load PV main effects are displayed in Figure 10. Significant main effects in PV occurred between different loads during the HC, JS, and HP exercises ($p < 0.001$). The greatest PV occurred at 30% 1RM HC (2.08 ± 0.40 m $\cdot$ s $^{-1}$). This was followed in order of magnitude by 45% (2.01 ± 0.33 m $\cdot$ s $^{-1}$), 65% (1.82 ± 0.21 m $\cdot$ s $^{-1}$), and 80% 1RM HC (1.69 ± 0.21 m $\cdot$ s $^{-1}$). Post hoc analysis revealed that PV at 30% 1RM HC was significantly greater than the PV produced at both 65% ($p < 0.001$, $d = 0.81$) and 80% 1RM HC ($p < 0.001$, $d = 1.22$). In addition, the PV produced at 45% 1RM HC was significantly greater than the PV produced at both 65% ($p < 0.001$, $d = 0.69$) and 80% 1RM HC ($p < 0.001$, $d = 1.16$). However, no significant difference in PV existed between 30 and 45% 1RM HC ($p = 0.059$, $d = 0.19$). Finally, 65% 1RM HC produced a significantly greater PV than 80% 1RM HC ($p < 0.001$, $d = 0.62$).

Exercise and Load Interaction

Significant interactions for PPO ($p < 0.001$), PF ($p = 0.014$), and PV ($p < 0.001$) were observed between the HC, JS, and HP

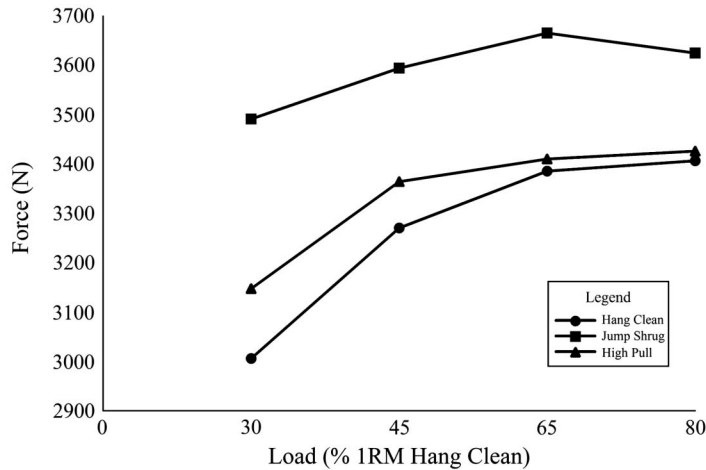


Figure 12. Exercise and load interaction for peak force ($p = 0.014$). 1RM = 1 repetition maximum.

exercises performed at different relative loads. The load at which the greatest PPO was produced, occurred at 65, 30, and 45% 1RM HC during the HC, JS, and HP, respectively (Figure 11). The exercise and load interactions for PF and PV are displayed in Figures 12 and 13, respectively. At every exercise load, the order of the greatest PPO, PF, and PV remained the same with the JS being the greatest followed in order by the HP and the HC. Given that the order of exercises remained the same (i.e., JS > HP > HC) in all variables measured throughout the loading spectrum within this study, the interaction seemed to be a result of the load.

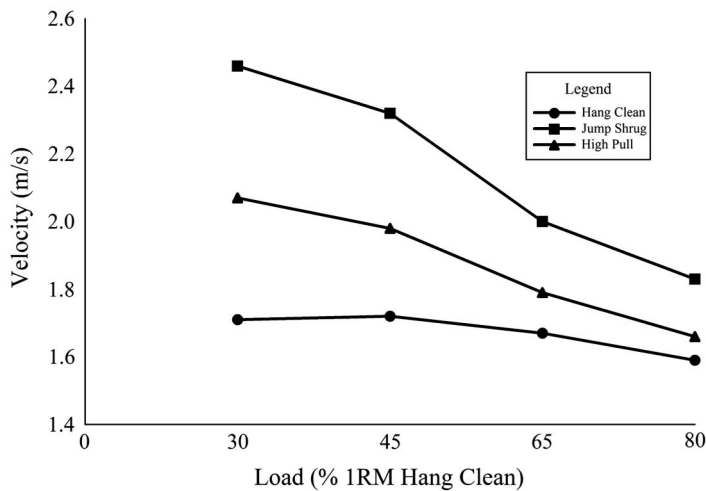


Figure 13. Exercise and load interaction for peak velocity ($p < 0.001$). 1RM = 1 repetition maximum.

Thus, the greatest differences in PPO, PF, and PV between the exercises occurred at the lighter loads (30 and 45% 1RM HC), but these differences were less observable at the heavier loads (65 and 80% 1RM HC).

DISCUSSION

It is likely that the ideal stimulus for improving muscular power involves training in a way where maximal power production is produced during sports-specific movements (24). The main purpose of this study was to compare the power production of the HC, JS, and HP when performed at different loads relative to the 1RM HC of each subject. The main find-

ings of this study were threefold. First, main effect differences in PPO existed between the HC, JS, and HP. The results of this study indicate that these differences are likely due to the PF and PV differences that existed between the exercises. For example, the JS produced significantly greater PF and PV than both the HC and HP, thus it makes sense that the JS produced the greatest PPO among the exercises examined. The secondary findings of this study included main effect differences in PPO, PF, and PV between the different exercise loads. Finally, there were interactions between the exercise (HC, JS, and HP) at specific loads for all of the variables

examined in the current study. As hypothesized, the JS produced the greatest PPO. These values were followed in order by the HP and HC. Also hypothesized, the PPO for the HC occurred at 65% 1RM HC.

Previous research has documented that success in sports seems to be strongly related to the ability of athletes to produce high levels of muscular power (1,3,4,7,8,10,11,13–15,18,20,21,23–26,28–31). The HC, JS, and HP variations of the power clean are vertical pull exercises that are used to train lower body power. The nature of these exercises is similar in that they are all dependent on a powerful shrug of the shoulders and triple extension. However, our results indicate

that the JS allowed for the greatest maximal power production by the subjects as compared with the HP and HC. In addition, the HP produced a greater PPO than the HC. It has been suggested that if athletes train using exercises that allow them to improve their muscular power, their overall athletic performance will also improve (23). Thus, it seems that the JS and HP exercises have the ability of producing high amounts of muscular power and should be considered as exercises that can be used to train lower body power.

Our results suggest that the largest contributing factor to the PPO of the JS and HP was the velocity of the lifter plus bar system during the movement. This seems logical because the JS and HP are more ballistic in nature than the HC. Our findings are supported by Newton et al. (29) who reported that the ballistic movement of a bench press throw, where the bar was released at the end of the range of motion, resulted in a greater velocity than a traditional bench press performed explosively. In the current study, part of the criterion for a successful repetition of the JS was that the subject's feet had to leave the platform during the movement as determined by observing the force-time curve immediately after the repetition. It is likely that this criterion required the subject's muscles to maintain higher force production throughout the entire range of motion, leading to a higher movement velocity (29). Therefore, the ability to produce a high velocity during the JS may be related to a greater need to focus on producing enough force and a fast enough velocity to leave the platform rather than focusing on catching the bar.

Although the HC is a highly beneficial exercise, it may be more time consuming to teach an athlete as compared with a power clean variation used to teach the HC (16). Previous authors recommend that practitioners should substitute less technical exercises to train lower body muscular power (1,20). By training with the JS and HP exercises, athletes with limited experience, injuries, or imperfect technique with the HC may still be able to effectively produce high levels of force, velocity, and power that seem to be important in sports performance. For example, if athletes struggle with HC technique, the results of this study indicate that it is possible to produce as much or greater lower body power by using the JS or HP as an alternative exercise. By implementing the JS or HP instead of the HC in this instance, there may be an increase in quality training time toward lower body muscular power, which will likely improve the athlete's overall performance in activities such as sprinting and jumping.

As previously mentioned, it has been suggested that strength and conditioning coaches should select exercises that allow their athletes to produce maximal power in the movement that is being trained (28). However, it is equally important for the practitioner to identify the loads that allow for maximal power production. Because many sports require high power output and explosiveness, it is preferred that athletes train at optimal loads so that the greatest stimulus for improved power output is provided (20,24,26,27,32). By training at the ideal load for each exercise, athletes will be

able to optimally improve their muscular power and, furthermore, their overall performance (23).

The main effects of load in the present study indicated that the subjects produced the greatest PPO at 45% 1RM HC. What this means to practitioners is that, in general, the HC, JS, and HP produced the highest PPO at a lower load as compared with a higher load. This is to be expected considering that the PPO of the JS and HP occurred at the lower loads of 30 and 45% 1RM HC, respectively. Because the load main effect combines all the repetitions performed by the subjects within the study, it may not be the best indicator of what load optimizes PPO. Previous research indicates that it may be important to prioritize the use of the optimal load with the exercise that allows the athlete to produce a high PPO (20,24,26,27).

In the present study, the PPO for the HC was found at 65% 1RM HC. This finding is supported by previous research that has indicated that the optimal load for the HC and power clean exercises occurred at either 70% (4,23) or 80% 1RM (5,6,9,25). However, it should be noted that several studies observed that there was no significant difference between the optimal load and 60–80% 1RM (4) or 50–90% 1RM (9,23,25), which makes the optimal load found within this study comparable to previous research. The PPO for the JS occurred at 30% 1RM HC. To our knowledge, this is the first study to assess and compare the optimal load for the JS, making it difficult to compare our results with previous research. Finally, the PPO for the HP in the current study occurred at 45% 1RM HC, which falls within the range of 30–60% 1RM previously noted for the HP by Thomas et al. (32).

Analysis of the interaction between the exercises and loads revealed that the greatest PPO, PF, and PV at each load were produced by the JS. This was followed in order by the HP and HC at all loads. The greatest differences in PPO, PF, and PV between the exercises were at the lighter loads of 30 and 45% 1RM HC. However, the differences between exercises were smaller at the heavier loads of 65 and 80% 1RM HC. In general, as the load increased, the amount of force increased for each exercise. Although this remained true throughout the loading spectrum for the HC and HP, this was not the case for the JS. As the load exceeded 65% 1RM HC during the JS, the magnitude of the force decreased, although still greater than both the HC and HP. This is likely due to the breakdown of technique during repetitions at 80% 1RM HC. As the load continued to increase for each exercise, the velocity of the lifter plus bar system decreased. Based on the force-velocity relationship, it makes sense that because each subject was getting closer to their 1RM load, it was difficult for them to produce a high velocity. In regard to the power output of each exercise, it should be noted that the power output of the JS dropped about 1000 W over the loading spectrum, whereas the HC and HP only differed about 500 and 600 W, respectively. This finding highlights the need for practitioners to prescribe

lighter loads for the JS exercise so that the athlete can effectively develop high levels of lower body muscular power.

A limitation of this study may have been the population selected. Athletic males with at least 2 years of previous experience with the HC were asked to participate in this study. That being said, no women, trained or untrained, were sought out as subjects. However, the subjects in the current study are part of a population that has been frequently examined throughout the literature and, therefore, this population may best allow for comparison with other studies. Finally, the current study used loads relative to the 1RM HC of each subject for each of the exercises examined. This was done to compare similar absolute loads in all exercises. The use of the array of loads (30–80%) was thought to be able to justify and identify differences of PF that may be apparent between exercises. Our results indicate that it is likely that the 1RM for the JS and HP are probably slightly higher than HC, so using a relative percentage of 1RM may have been another way of comparing loading; however, performing a 1RM test for the JS and HP in an athletic setting may not be practical.

The true optimal load for each exercise may be similar to those reported within this study, but it is recommended that future research may consider the use of smaller loading increments to better determine the optimal load for each exercise. Future research may consider the use of different populations, such as untrained men and both trained and untrained women, although comparing the HC, JS, and HP. To accurately determine how well each of these exercises trains lower body power, future research should consider analyzing the HC, JS, and HP using 3-dimensional motion analysis equipment to compare to what extent the hip, knee, and ankle joints extend during the second pull movement.

PRACTICAL APPLICATIONS

The results of this study may assist strength and conditioning practitioners in selecting exercises that maximize lower body power production during training, which may then enhance an athlete's performance in their respective sports. Because the JS and HP variations of the power clean exercise were superior to the HC in producing power, force, and velocity of the lifter plus bar system over the entire range of loads examined, it is suggested that strength and conditioning practitioners consider implementing the JS and HP exercises into their training regimens, especially for those who have trouble learning the HC. The JS and HP can be used as primary methods to improve lower body muscular power, but should also be used to complement exercises that are already being utilized to improve lower body muscular strength and power. To optimize power production with both the JS and HP exercises, practitioners should consider using loads at ~30 and 45% of each athlete's 1RM HC, respectively.

ACKNOWLEDGMENTS

This study was supported by a grant from the University of Wisconsin-La Crosse in La Crosse, Wisconsin. The authors

would like to sincerely thank the subjects who participated in this research and made this project possible. The results of this study do not constitute endorsement of the product by the authors or the National Strength and Conditioning Association. There is no conflict of interest. There are no professional relationships with companies or manufacturers who will benefit from the results of the present study for each author.

REFERENCES

1. Baker, D and Nance, S. The relation between strength and power in professional rugby league players. *J Strength Cond Res* 13: 224–229, 1999.
2. Comfort, P, Allen, M, and Graham-Smith, P. Comparisons of peak ground reaction force and rate of force development during variations of the power clean. *J Strength Cond Res* 25: 1235–1239, 2011.
3. Comfort, P, Allen, M, and Graham-Smith, P. Kinetic comparisons during variations of the power clean. *J Strength Cond Res* 25: 3269–3273, 2011.
4. Comfort, P, Fletcher, C, and McMahon, JJ. Determination of optimal load during the power clean in collegiate athletes. *J Strength Cond Res* 26: 2970–2974, 2012.
5. Cormie, P, McBride, JM, and McCaulley, GO. The influence of body mass on calculation of power during lower-body resistance exercises. *J Strength Cond Res* 21: 1042–1049, 2007.
6. Cormie, P, McBride, JM, and McCaulley, GO. Validation of power measurement techniques in dynamic lower body resistance exercises. *J Appl Biomech* 23: 103–118, 2007.
7. Cormie, P, McBride, JM, and McCaulley, GO. Power-time, force-time, and velocity-time curve analysis during the jump squat: Impact of load. *J Appl Biomech* 24: 112–120, 2008.
8. Cormie, P, McCaulley, GO, and McBride, JM. Power versus strength-power jump squat training: Influence on the load-power relationship. *Med Sci Sports Exerc* 39: 996–1003, 2007.
9. Cormie, P, McCaulley, GO, Triplett, NT, and McBride, JM. Optimal loading for maximal power output during lower-body resistance exercises. *Med Sci Sports Exerc* 39: 340–349, 2007.
10. Cronin, J, McNair, PJ, and Marshall, RN. Developing explosive power: A comparison of technique and training. *J Sci Med Sport* 4: 59–70, 2001.
11. Garhammer, J. Power production by Olympic weightlifters. *Med Sci Sports Exerc* 12: 54–60, 1980.
12. Garhammer, J. Power clean: Kinesiological evaluation. *Natl Strength Cond Assoc J* 6: 61–63, 1984.
13. Garhammer, J. A review of power output studies of Olympic and powerlifting: Methodology, performance, prediction, and evaluation tests. *J Strength Cond Res* 7: 76–89, 1993.
14. Gentry, RM. A comparison of two instructional methods of teaching the power clean weight training exercise to intercollegiate football players with novice power clean experience. Doctoral dissertation, Virginia Polytechnic Institute and State University, 1999.
15. Hardee, JP, Triplett, NT, Utter, AC, Zwetsloot, KA, and McBride, JM. Effect of interrepetition rest on power output in the power clean. *J Strength Cond Res* 26: 883–889, 2012.
16. Hedrick, A. Teaching the clean. *Strength Cond J* 26: 70–72, 2004.
17. Hopkins, WG. A scale of magnitudes for effect statistics. Available at: <http://sportsjournal.org/resource/stats/effectmag.html>. Accessed January 12, 2013.
18. Hori, N, Newton, RU, Andrews, WA, Kawamori, N, McGuigan, MR, and Nosaka, K. Comparison of four different

- methods to measure power output during the hang power clean and the weighted jump squat. *J Strength Cond Res* 21: 314–320, 2007.
19. Hori, N, Newton, RU, Nosaka, K, and McGuigan, MR. Comparison of different methods of determining power output in weightlifting exercises. *Strength Cond J* 28: 34–40, 2006.
 20. Hori, N, Newton, RU, Nosaka, K, and Stone, MH. Weightlifting exercises enhance athletic performance that requires high-load speed strength. *Strength Cond J* 27: 50–55, 2005.
 21. Hydock, D. The weightlifting pull in power development. *Strength Cond J* 23: 32–37, 2001.
 22. Kaneko, M, Fuchimoto, T, Toji, H, and Suei, K. Training effect of different loads on the force-velocity relationship and mechanical power output in human muscle. *Scand J Sport Sci* 5: 50–55, 1983.
 23. Kawamori, N, Crum, AJ, Blumert, PA, Kulik, JR, Childers, JT, Wood, JA, Stone, MH, and Haff, GG. Influence of different relative intensities on power output during the hang power clean: Identification of the optimal load. *J Strength Cond Res* 19: 698–708, 2005.
 24. Kawamori, N and Haff, GG. The optimal training load for the development of muscular power. *J Strength Cond Res* 18: 675–684, 2004.
 25. Kilduff, LP, Bevan, H, Owen, N, Kingsley, MIC, Bunce, P, Bennett, M, and Cunningham, D. Optimal loading for peak power output during the hang power clean in professional rugby players. *Int J Sports Physiol Perform* 2: 260–269, 2007.
 26. Kirby, TJ, Erickson, T, and McBride, JM. Model for progression of strength, power, and speed training. *Strength Cond J* 32: 86–90, 2010.
 27. Moss, BM, Refsnes, PE, Abildgaard, A, Nicolaysen, K, and Jensen, J. Effects of maximal effort strength training with different loads on dynamic strength, cross-sectional area, load-power and load-velocity relationships. *Eur J Appl Physiol Occup Physiol* 75: 193–199, 1997.
 28. Newton, RU and Kraemer, WJ. Developing explosive muscular power: Implications for a mixed methods training strategy. *Strength Cond J* 16: 20–31, 1994.
 29. Newton, RU, Kraemer, WJ, Hakkinen, K, Humphries, BJ, and Murphy, AJ. Kinematics, kinetics, and muscle activation during explosive upper body movements. *J Appl Biomech* 12: 31–43, 1996.
 30. Stone, MH, O'Bryant, HS, McCoy, L, Coglianese, R, Lehmkuhl, M, and Schilling, B. Power and maximum strength relationships during performance of dynamic and static weighted jumps. *J Strength Cond Res* 17: 140–147, 2003.
 31. Takano, B. The power clean-perspectives and preparations. *Natl Strength Cond Assoc J* 14: 68–71, 1992.
 32. Thomas, GA, Kraemer, WJ, Spiering, BA, Volek, JS, Anderson, JM, and Maresh, CM. Maximal power at different percentages of one repetition maximum: Influence of resistance and gender. *J Strength Cond Res* 21: 336–342, 2007.
 33. Winchester, JB, Erickson, TM, Blaak, JB, and McBride, JM. Changes in bar-path kinematics and kinetics after power-clean training. *J Strength Cond Res* 19: 177–183, 2005.